



SEGMENTATION METHODS – WATERSHED AND EDGE-BASED

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WATERSHED

When using the watershed segmentation algorithm, there are a few parameters we can control to affect the outcome of the algorithm. These are the parameters:

- Kernel Size – reflects the number of pixels examined when blurring each individual pixel during a gaussian blur.
- Opening Iterations – Reflects the number of iterations when applying morphological opening to the image. (Iterations of erosion on the image to remove noise)
- Dilation Iterations – Reflects the number of iterations when applying dilation to the eroded image. (Iterations of dilation to fill gaps and smooth boundaries)
- Distance Transform Multiplier – Specifies how far a pixel must be from a sure background pixel to be considered a foreground pixel. (Higher multiplier means fewer sure foreground pixels)

The next slides will show the results of changing these parameters. The results will show the original image, the image after the watershed algorithm runs, the calculated sure background of the image, and the calculated sure foreground of the image.

KERNEL SIZE

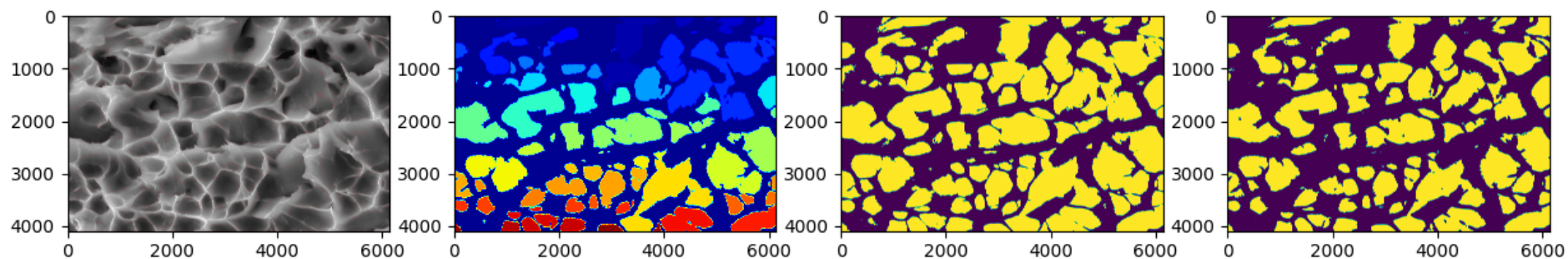
Original

Watershed

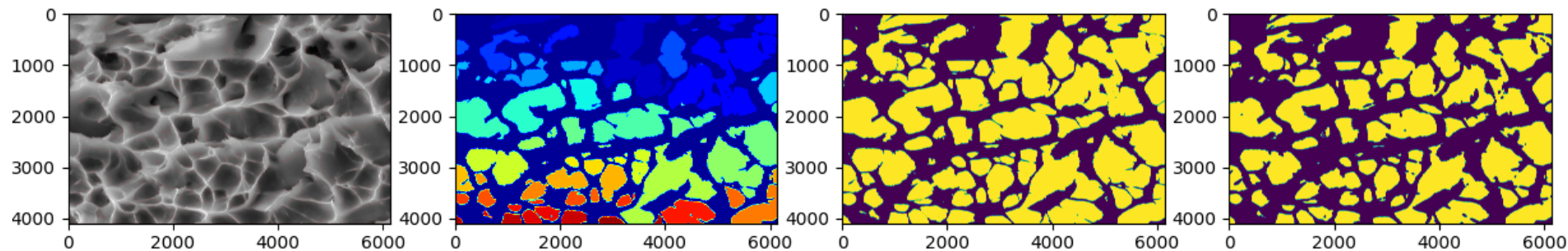
Background

Foreground

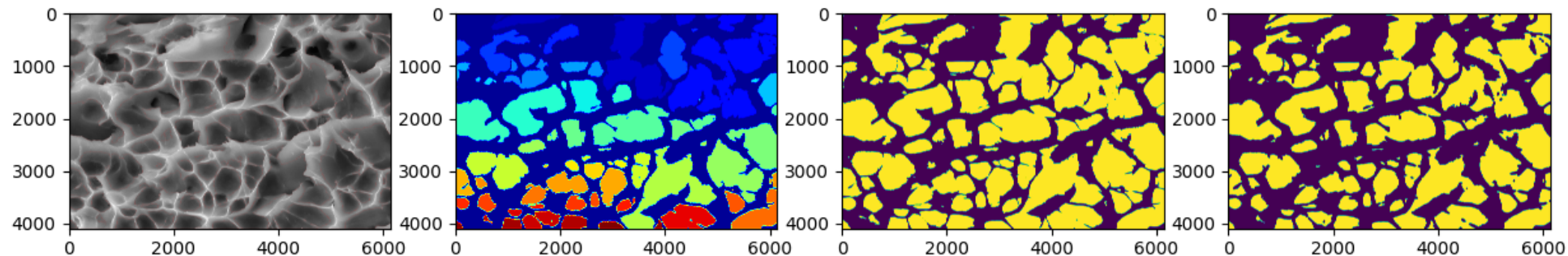
3x3 Kernel Grid:



9x9 Kernel Grid:



11x11 Kernel Grid:



KERNEL SIZE - CONTINUED

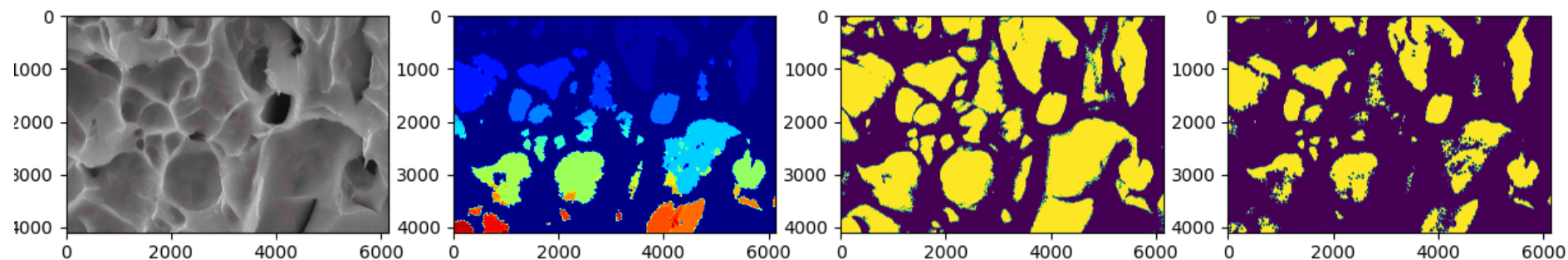
Original

Watershed

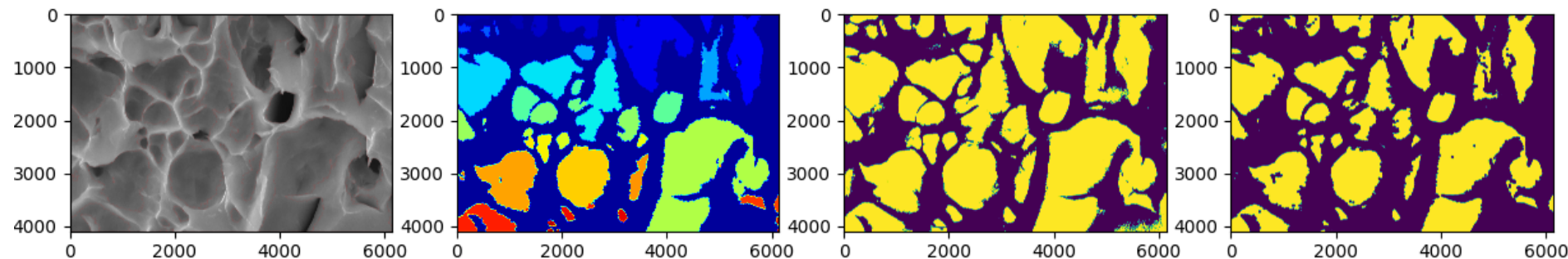
Background

Foreground

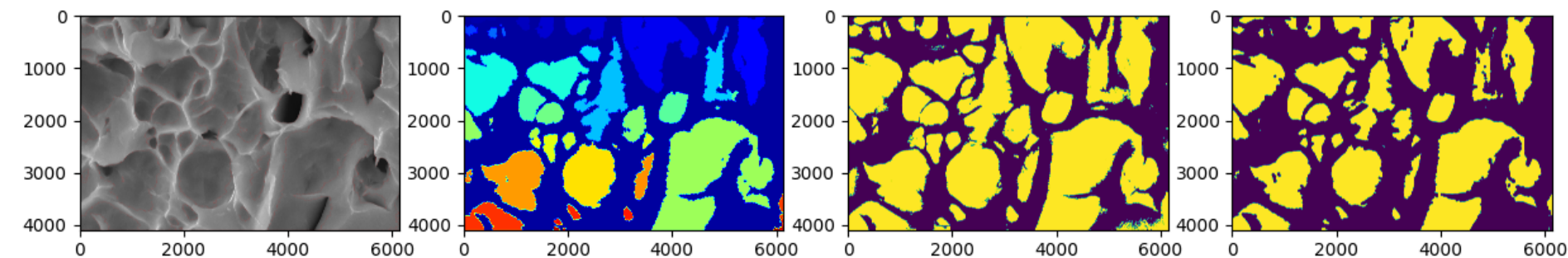
3x3 Kernel Grid:



9x9 Kernel Grid:



11x11 Kernel Grid:



KERNEL SIZE - CONCLUSION

After experimenting with increasing the kernel size from 3, the minimum value, it appears that larger kernel sizes perform better. With the first image used, the changes were negligible, but with the second image used, the algorithm worked significantly better as the kernel size increased. This helps to know that increasing the kernel size for smoothing can improve the effectiveness of watershed on more complex pore images.

OPENING ITERATIONS

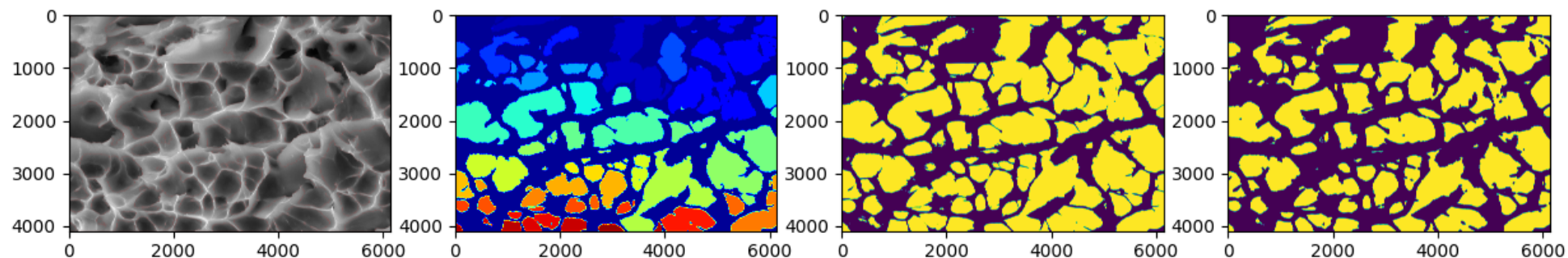
Original

Watershed

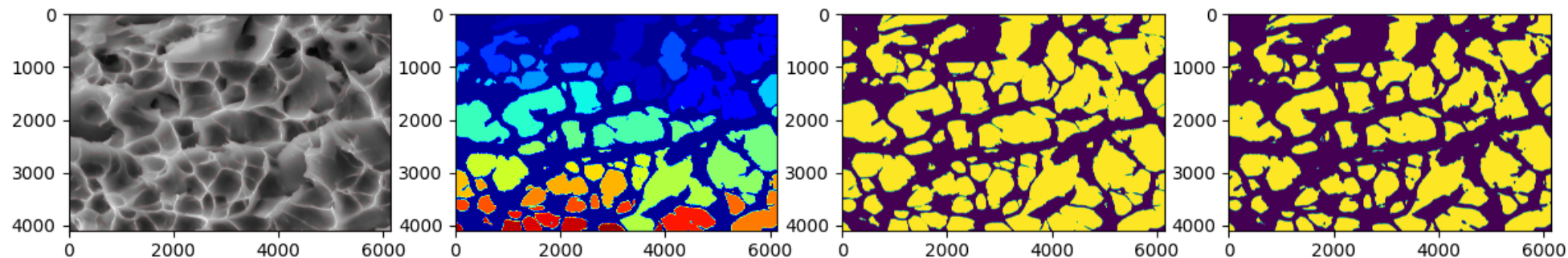
Background

Foreground

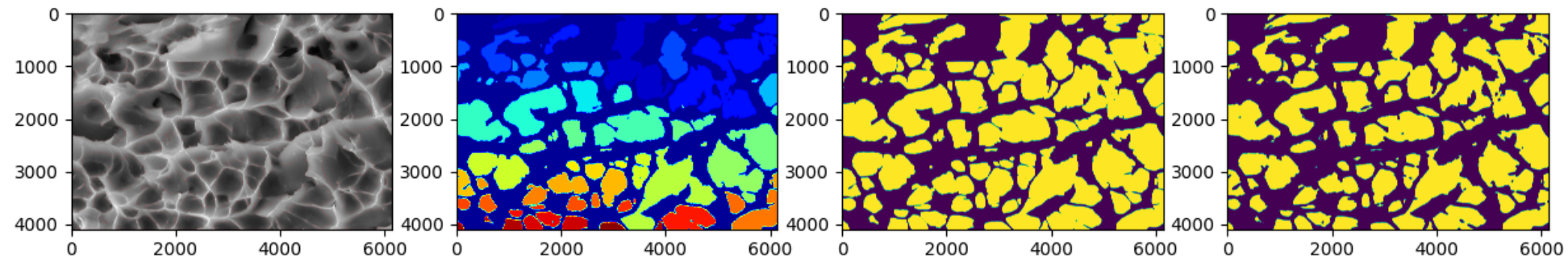
1 Iteration:



5 Iterations:

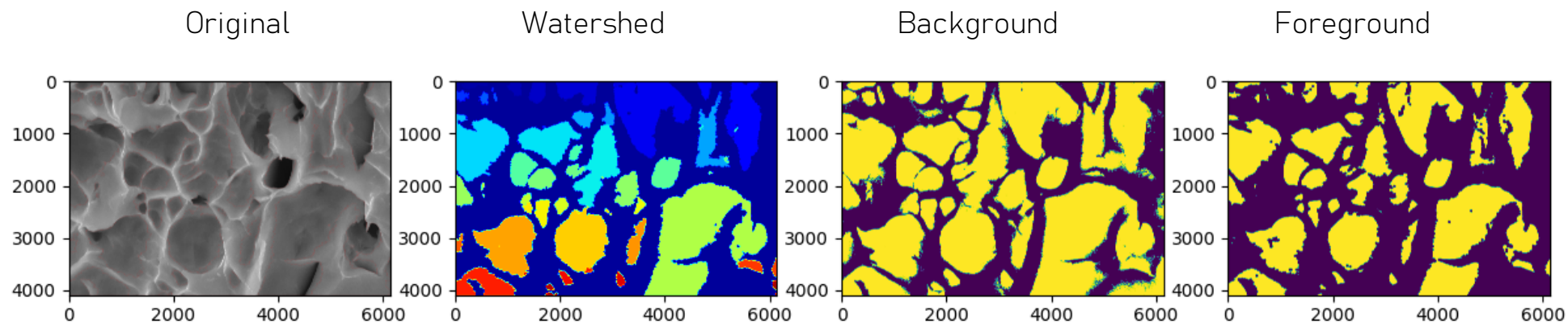


15 Iterations:

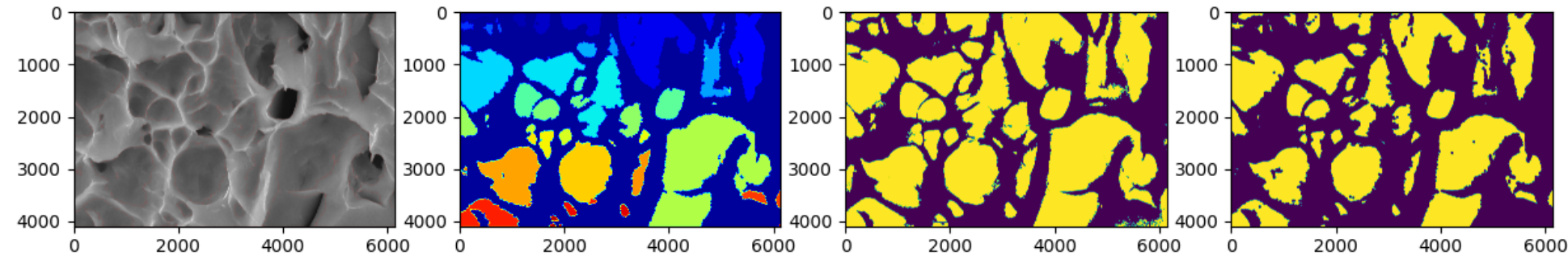


OPENING ITERATIONS - CONTINUED

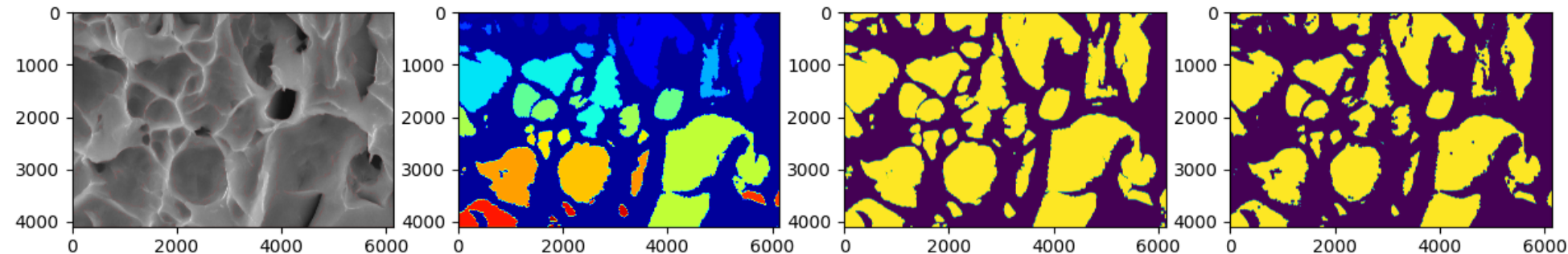
1 Iteration:



5 Iterations:



15 Iterations:



OPENING ITERATIONS - CONCLUSION

From increasing the amount of iterations, I noticed that there were no significant changes until 15 iterations were tested. In that case, there were differences shown in the noise levels of the background and foreground images. These iterations go hand in hand with the Dilation Iterations, so it is likely that this noise will have a significant effect on how the further iterations operate.

DILATION ITERATIONS

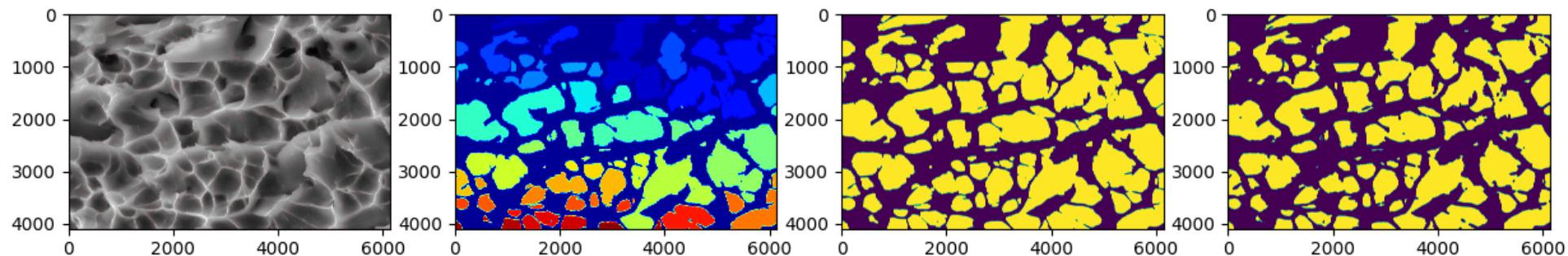
Original

Watershed

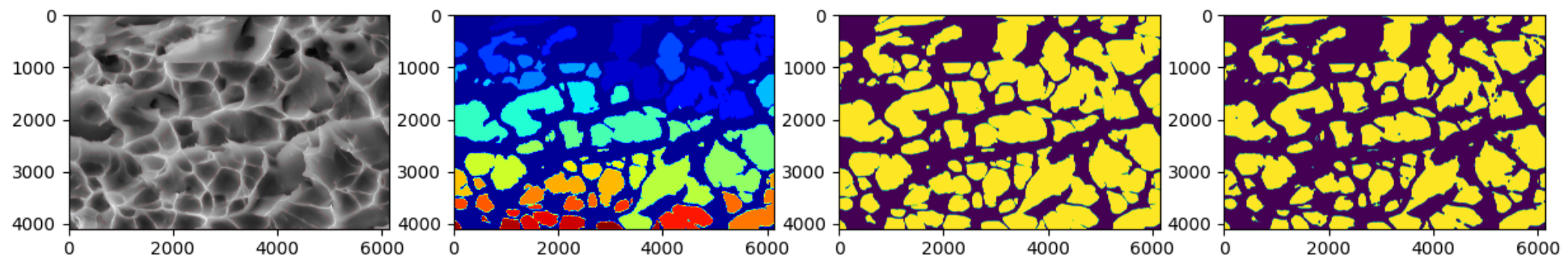
Background

Foreground

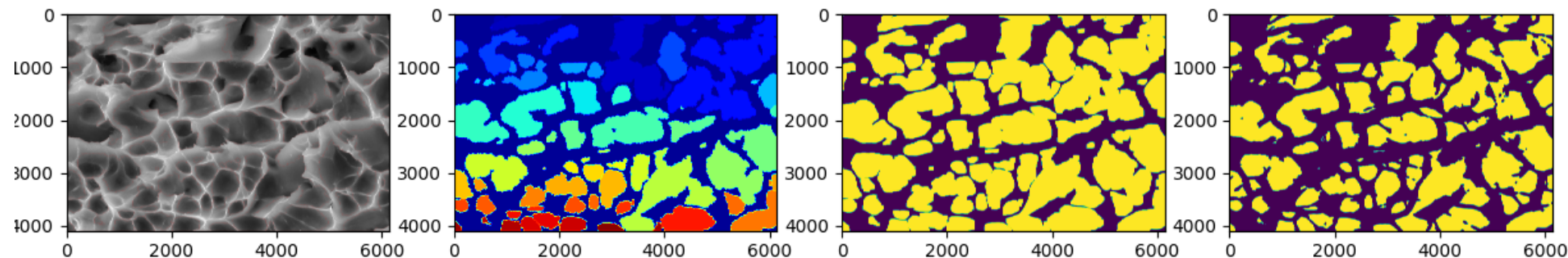
1 Iteration:



5 Iterations:



15 Iterations:



DILATION ITERATIONS - CONTINUED

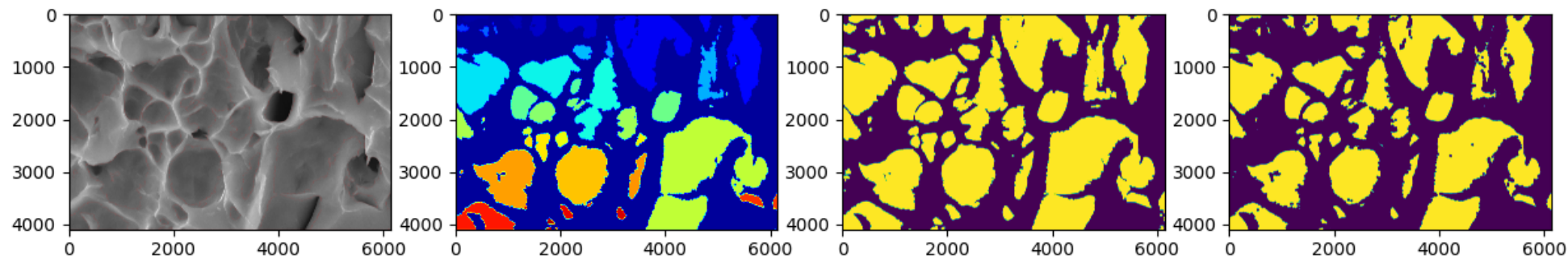
Original

Watershed

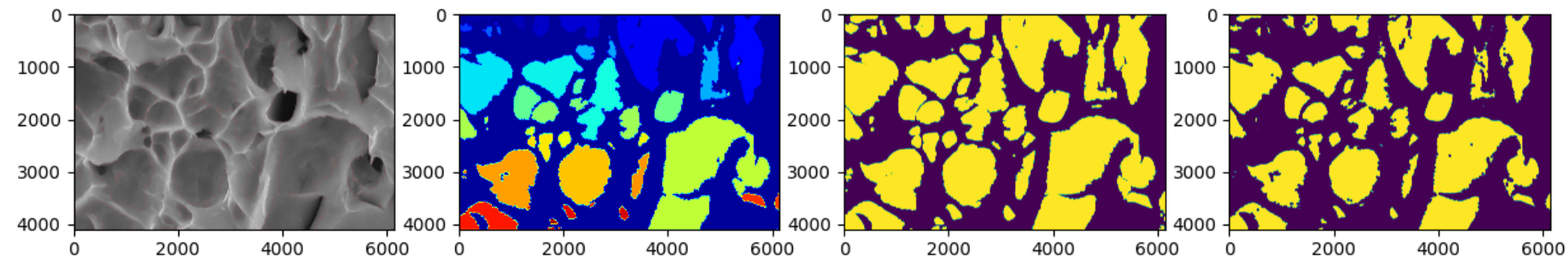
Background

Foreground

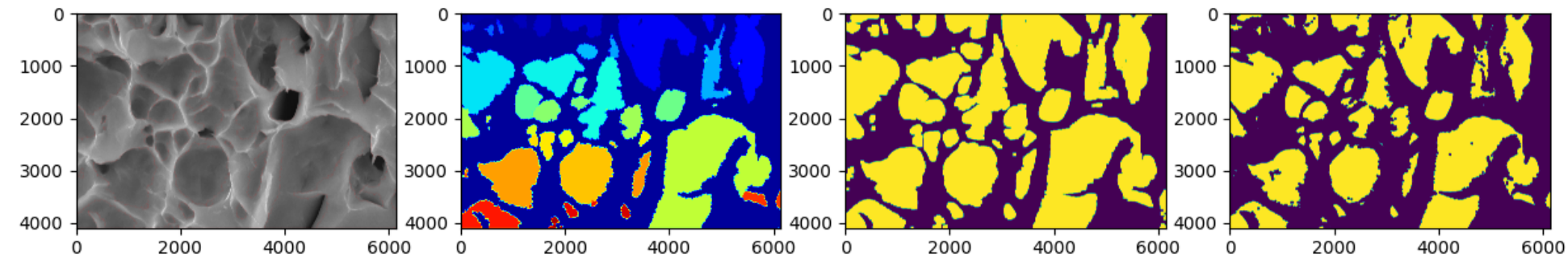
1 Iteration:



5 Iterations:



15 Iterations:



DILATION ITERATIONS - CONCLUSION

Increasing the number of Dilation Iterations has shown that holes or gaps that appeared as a result of Opening Iterations will close up. These parameter tests were performed with the Opening Iterations value set to 15 in an attempt to show more diverse results.

DISTANCE TRANSFORM MULTIPLIER

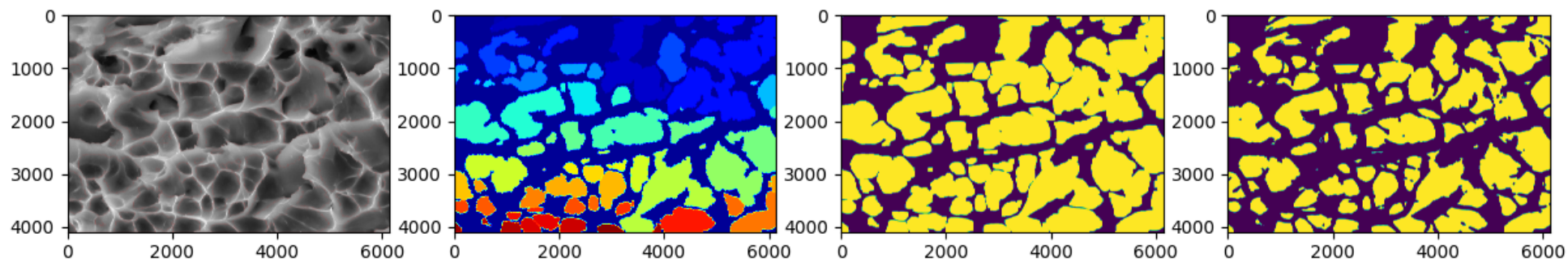
Original

Watershed

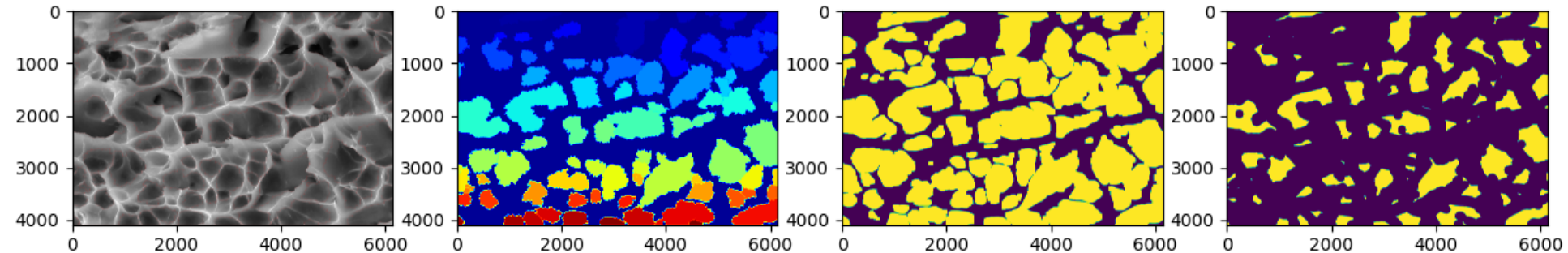
Background

Foreground

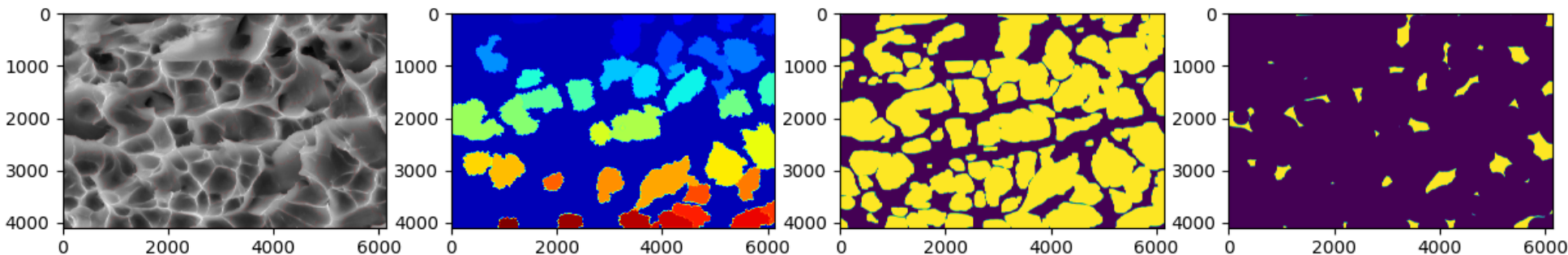
0.05 Multiplier:



0.25 Multiplier:



0.5 Multiplier:



DISTANCE TRANSFORM MULTIPLIER - CONTINUED

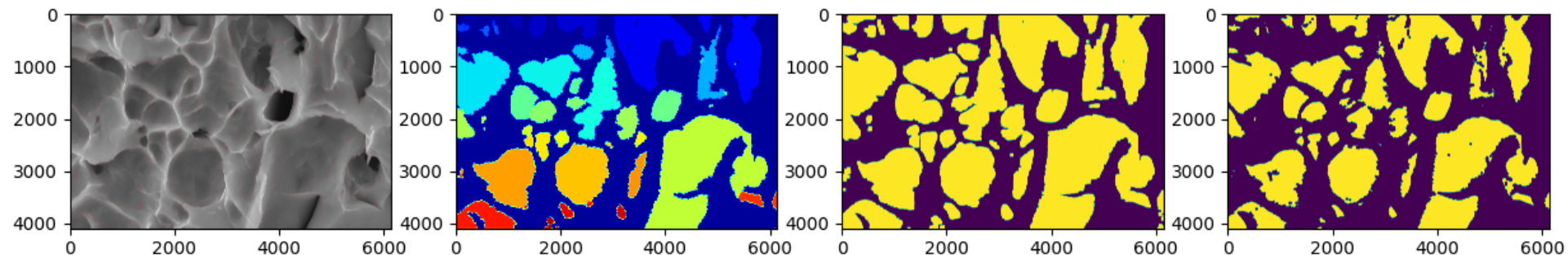
Original

Watershed

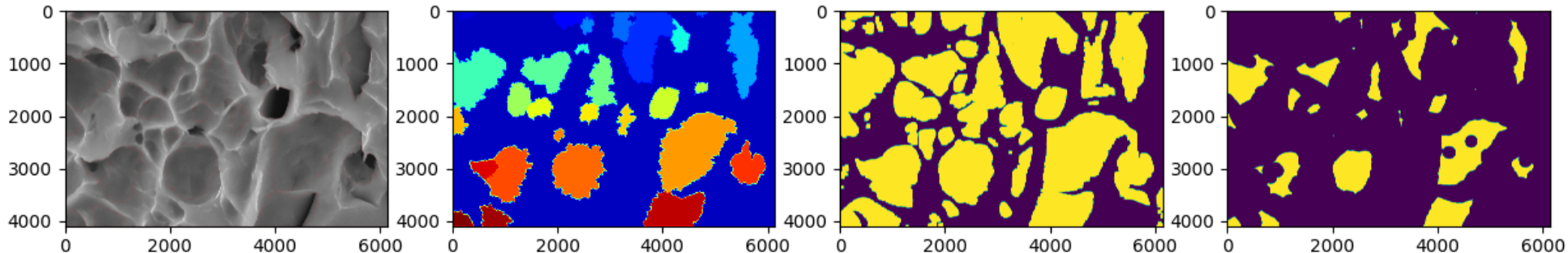
Background

Foreground

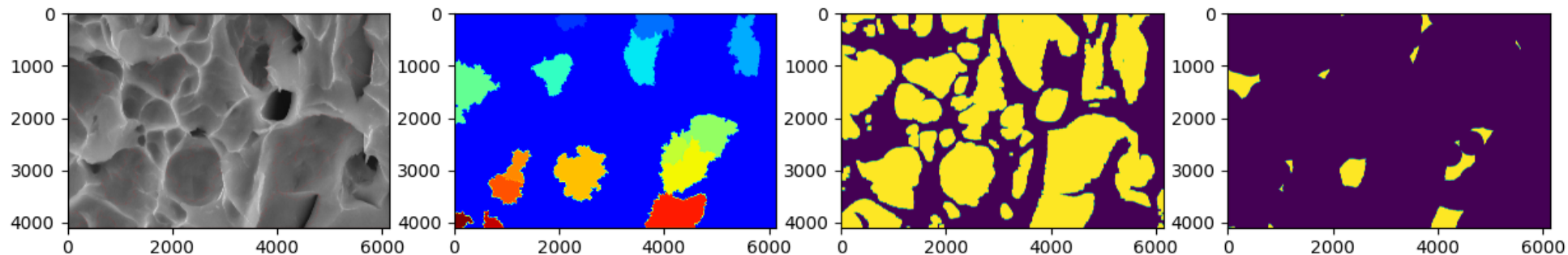
0.05 Multiplier:



0.25 Multiplier:



0.5 Multiplier:



DISTANCE TRANSFORM MULTIPLIER - CONCLUSION

Increasing the distance transform multiplier showed that the amount of pixels deemed to be sure foreground was greatly reduced. Due to the high frequency of smaller pores in the images, it seems that a smaller multiplier will work much better.

WATERSHED - CONCLUSION

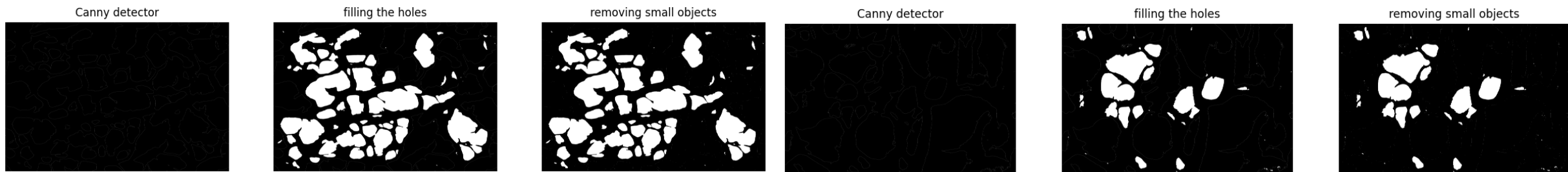
After experimenting with the ranges of different parameters, we can conclude that watershed is a viable option as a segmentation algorithm. With further tuning and testing, the results can be optimized to produce better and more accurate results.

EDGE-BASED SEGMENTATION

Edge-based segmentation did not prove to be useful after some testing. The tunable parameters are:

- Kernel Size – reflects the number of pixels examined when blurring each individual pixel during a gaussian blur.
- Minimum Object Size – reflects the size of the smallest object in the image, smaller objects are considered to be noise.

After manipulating these parameters, the best achievable results can be seen below:



EDGE-BASED SEGMENTATION - CONCLUSION

As you can see from the images, some of the pores are properly segmented, but a lot of pores were not detected. This is due to the edges not being noticed entirely, resulting in some of the pores not having full borders.

WATERSHED – COMPARISON WITH GROUND TRUTH

